$A \cdot B = C$

$(m,n)\ (n,p) \qquad (m,p)$

R: $(\lambda+\mu)\eta = \lambda\eta + \mu\eta$

$\lambda + (\mu\eta) \neq (\lambda+\mu)\cdot(\lambda+\eta)$

SV: V $v+w$

$\lambda \cdot v$

Exemple

1) $V = K^n$ — $(V, +, 0)$ grup comutativ

2) $V = M_{m,n}(K)$

3) $V = Z_p$ $(p \text{ prim})$ $K = Z_p$ $v_n = 1$

$v = (x) \qquad v = x$

$p = 2 \quad K = Z_2 \quad V = Z_2^m$

$v = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$

$v + w = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

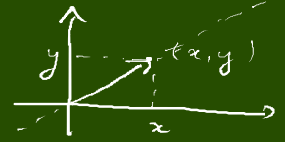
$0 \cdot v = 0 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$1 \cdot v = 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$

4) $V = \mathbb{C}$ $\quad k = \mathbb{R}$

$\lambda \in \mathbb{R}$ $\quad z \in \mathbb{C}$ $\quad z = x + iy$

$\lambda z = \lambda(x+iy) = (\lambda x) + i(\lambda y)$



$\mathbb{R}^2$

$V = \mathbb{C}$ $\quad k = \mathbb{C}$

$\lambda \in \mathbb{C}$ $\quad z \in \mathbb{C}$ $\quad z = x + iy$

$\lambda = \alpha + i\beta$

$\lambda z = (\alpha + i\beta)(x + iy) =$

$= (\alpha x - \beta y) + i(\alpha y + \beta x)$

$0_1 \neq 0_2$

$(V, +, 0_1)$ $\quad$ $(V, +, 0_2)$

$\forall v \in V \quad v + 0_1 = 0_1 + v = v \quad (1)$

$\forall v \in V \quad v + 0_2 = 0_2 + v = v \quad (2)$

(1) $v = 0_2 \quad 0_2 + 0_1 = 0_1 + 0_2 = 0_2$

(2) $v = 0_1 \quad 0_1 + 0_2 = 0_2 + 0_1 = 0_1$

$\Rightarrow 0_1 = 0_2$

$0_k \cdot v = 0_V$

k → scalar; V → vector

$k: \quad 0_k + 0_k = 0_k$

$$O_k \cdot v = (O_k + O_k) \cdot v = (O_k \cdot v) + (O_k \cdot v) \Big| + (-(O_k \cdot v))$$

$$\underbrace{(O_k \cdot v) + (-(O_k \cdot v))} = ((O_k \cdot v) + (O_k \cdot v)) + (-(O_k \cdot v))$$

$$O_V = (O_k \cdot v) + \underbrace{((O_k \cdot v) + (-(O_k \cdot v)))}_{O_V}$$

$$O_V = (O_k \cdot v) + O_V = O_k \cdot v$$

k

$n = \text{grad } P$

$$k[x] = \{ P(x) = a_0 + a_1 x + \dots + a_n x^n \mid a_0, \dots, a_n \in k, a_n \neq 0 \}$$

es.: $P(x) = 1 - 2x + 3x^2 \in \mathbb{R}[x]$

$$Q(x) = i + \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)x^3 \in \mathbb{C}[x]$$

$$P(x) = a_0 + a_1 x + \dots + a_n x^n \in k[x]$$

$$Q(x) = b_0 + b_1 x + \dots + b_m x^m \in k[x]$$

$$P(x) + Q(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots$$

$P(x) = 1 - 2x + 3x^2$ $\quad P(x) = 0$

$Q(x) = 4 - x^2 + 5x^4$

$P(x) + Q(x) = 5 - 2x + 2x^2 + 5x^4$

$(k[x], +, 0)$ gr. com.

$\rightarrow -P(x) = -1 + 2x - 3x^2$

$\lambda \in k \qquad P(x) \in k[x]$

$P(x) = a_0 + a_1 x + \ldots + a_n x^4$

$\lambda \cdot P(x) = (\lambda a_0) + (\lambda a_1) x + \ldots + (\lambda a_n) x^4$

$k = \mathbb{R}$

$M = \mathbb{R}$ $\qquad f : \mathbb{R} \to \mathbb{R}$

f, g

$f + g : \mathbb{R} \to \mathbb{R}$

$(f + g)(x) = f(x) + g(x)$

$(\lambda \cdot f)(x) = \lambda \cdot f(x)$

$\lambda \cdot f : \mathbb{R} \to \mathbb{R}$, $\lambda \in \mathbb{R}$

$k = \mathbb{R} \qquad M = \mathbb{N}$

$f : \mathbb{N} \longrightarrow \mathbb{R}$ sono

$f(0), f(1), f(2), \ldots, f(n), \ldots$

$\{ f : \mathbb{N} \longrightarrow \mathbb{R} \mid f \text{ qlc} \} =$

$= \{ (a_n)_{n \geq 0} \mid a_n \in \mathbb{R} \} = \mathbb{R}^{\infty}$

$\mathbb{R}[x] = \{ P(x) = a_0 + \ldots + a_n x^n \mid a_n \in \mathbb{R} \}$

$\{$

$(a_0, a_1, \ldots, a_n, 0, 0, 0, \ldots)$

$\underbrace{\lambda v + \mu w}$

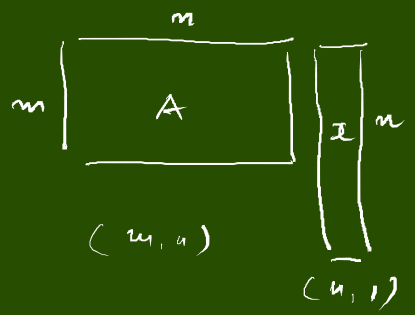
combinație liniară

$\mu = 0$: $\lambda \in k, v \in U \Rightarrow \underline{\lambda v \in U}$

$\lambda = \mu = 1$: $v, w \in U \Rightarrow \underline{v + w \in U}$

$A \in \mathcal{M}_{m,n}(k)$

$\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$



$= 0_m$ vector

$(m, 1)$

$\lambda, \mu \in K$
$x, y \in Ker(A)$ $\Big| \overset{?}{\Rightarrow} \lambda x + \mu y \in Ker(A)$

$x \in Ker(A) \iff Ax = 0_m$
$y \in Ker(A) \iff Ay = 0_m$

$$A(\lambda x + \mu y) = \lambda \underbrace{(Ax)}_{0_m} + \mu \cdot \underbrace{(Ay)}_{0_m} = 0_m$$

$\Rightarrow \lambda x + \mu y \in Ker(A)$

$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ $\quad Ker(A) = ?$

$$Ker(A) = \left\{ v = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \,\middle|\, Av = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} =$$

$$= \left\{ v \in \mathbb{R}^2 \,\middle|\, \begin{cases} 2x + 3y = 0 \\ x + 4y = 0 \end{cases} \right\}$$

$$= \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

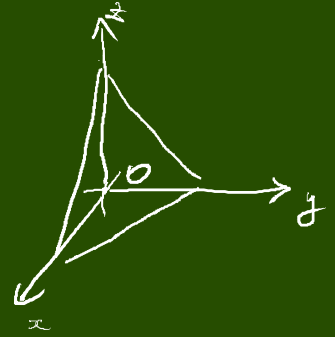
$A = \begin{pmatrix} 0 & 0 \end{pmatrix}$

$$\ker(A) = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid A \begin{pmatrix} x \\ y \end{pmatrix} = 0 \right\}$$

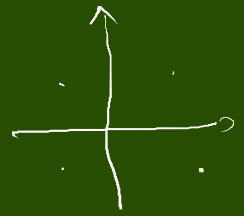
$$= \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid 0 \cdot x + 0 \cdot y = 0 \right\}$$

$$= \mathbb{R}^2$$

$\{0\} \subseteq$ O $\in$ dreapta $\subseteq \mathbb{R}^2$



$ax + by + cz + d = 0$



$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \qquad \ker(A) = ?$$

$(2,3)$ $(3,1)$

$$\ker(A) = \left\{ v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; Av = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$\begin{cases} x - 2y + z = 0 \\ y + 2z = 0 \end{cases} \qquad \begin{array}{l} \Rightarrow x = -4z - z = -5z \\ \Rightarrow y = -2z \end{array}$$

$z \in \mathbb{R}$

$$\ker(A) = \left\{ v = \begin{pmatrix} -5z \\ -2z \\ z \end{pmatrix} \;\middle|\; z \in \mathbb{R} \right\}$$

$$= z \begin{pmatrix} -5 \\ -2 \\ 1 \end{pmatrix}$$

$\pi_1 : \quad x - 2y + z = 0$

$\pi_2 : \quad y + 2z = 0$

$$A = \begin{pmatrix} a & b & c \\ \lambda a & \lambda b & \lambda c \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$\begin{cases} ax + by + cz = 0 \\ \lambda a x + \lambda b y + \lambda c z = 0 \end{cases}$$